

SEMAI: predicting greenhouse misting two hours ahead

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SDG 9

INDUSTRY, INNOVATION AND
INFRASTRUCTURE

PSC GET Talent 2026 · Machine Learning (software)

PROBLEM STATEMENT

Plants lose water according to **VPD** (vapour-pressure deficit, kPa), which depends on heat as well as humidity. Our tabletop greenhouse mists when VPD rises above **1.2 kPa**, but a controller only reacts once the air is already dry.

SEMAI Forecast uses supervised machine learning to warn **two hours ahead**, so small growers in Penang can prepare before the hot afternoon.

66.9%

of Penang's dry-stress hours (2,379 of 3,555) are missed by the simple rule "mist when humidity drops below 60 %". A typical missed hour: 30.1 °C and 66 % humidity, yet VPD is 1.42 kPa.

DATA SOURCE

Open-Meteo Historical Weather API (ERA5 reanalysis), downloaded as a raw CSV.

- Penang, 5.45° N 100.29° E · hourly, 1 Jan 2023 – 31 Dec 2025
- 26,304 rows × 26 columns** (25 weather variables)
- Licence CC BY 4.0 · real data only, nothing synthetic

Cleaning: removed metadata rows and units from the headers; checked missing values (O), duplicate hours (O), skipped hours (O) and physical ranges (O out of range); dropped 4 redundant columns (rain = precipitation, sea-level pressure, is-day, weather code); encoded wind direction as sine and cosine.

VARIABLES

Target (classification): *mist_next_2h* = 1 if VPD > 1.2 kPa at hour t+1 or t+2 (16.1 % of hours).

FROM THE DATASET (20)

- Temperature, humidity, dew point, apparent temperature
- VPD now
- Shortwave, direct, diffuse radiation; sunshine
- Cloud cover: total, low, mid, high
- Wind speed, gusts
- Precipitation, evapotranspiration (ET_o)
- Soil temperature, soil moisture
- Surface pressure

ENGINEERED (9)

- Hour of day (sin, cos)
- Day of year (sin, cos)
- Wind direction (sin, cos)
- 1-hour change in VPD
- 1-hour change in temperature
- 1-hour change in radiation

PREDICTION MODEL (GRADIENT BOOSTING)

A **gradient-boosted decision-tree classifier** (scikit-learn *HistGradientBoostingClassifier*) gives, for every hour, the probability that misting will be needed within 2 hours. Trained on **2023–2024** (17,541 hours), tested on **all of 2025** (8,758 hours), which the model never saw.

94.5%

accuracy on unseen
2025

0.885

F1 score
(rule: 0.773)

87.2%

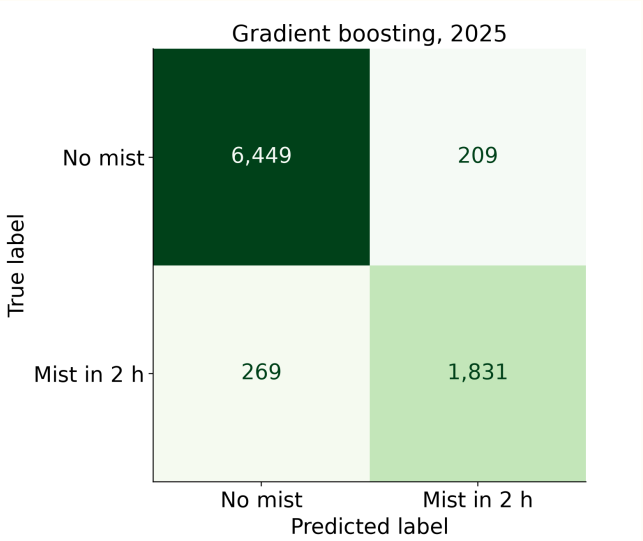
misting hours caught
(1,831 of 2,100)

0.985

ROC AUC

MODEL EVALUATION

METHOD (2025 TEST)	ACC.	PREC.	RECALL	F1
Always "no"	76.0%	0.000	0.000	0.000
"Dry right now" rule	89.9%	0.839	0.717	0.773
Usual dry hours rule	83.9%	0.635	0.772	0.697
Logistic regression	94.2%	0.881	0.879	0.880
Gradient boosting	94.5%	0.898	0.872	0.885



FEATURE SELECTION

TOP 5 FEATURES (DROP IN F1 WHEN SHUFFLED)

VPD now	0.287
Hour of day (cos)	0.034
Soil moisture, 0–7 cm	0.026
1-hour change in sunlight	0.022
Relative humidity	0.020

Permutation importance on the 2025 test set: how much F1 drops when one feature is shuffled.

VPD now: dry air rarely turns humid within an hour, and VPD is what drives plant water stress (Grossiord et al., 2020).

Hour of day: the sun's daily cycle; radiation drives evaporation and heating (Allen et al., 1998).

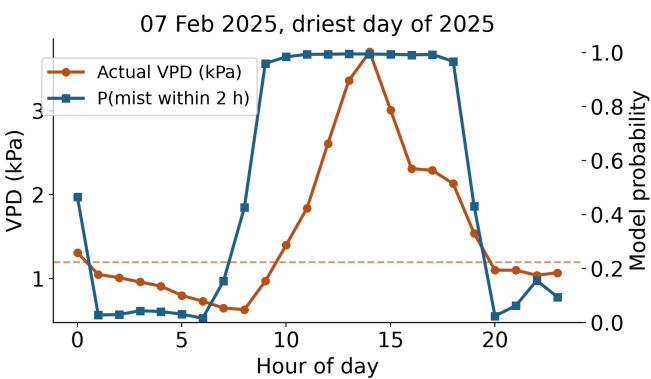
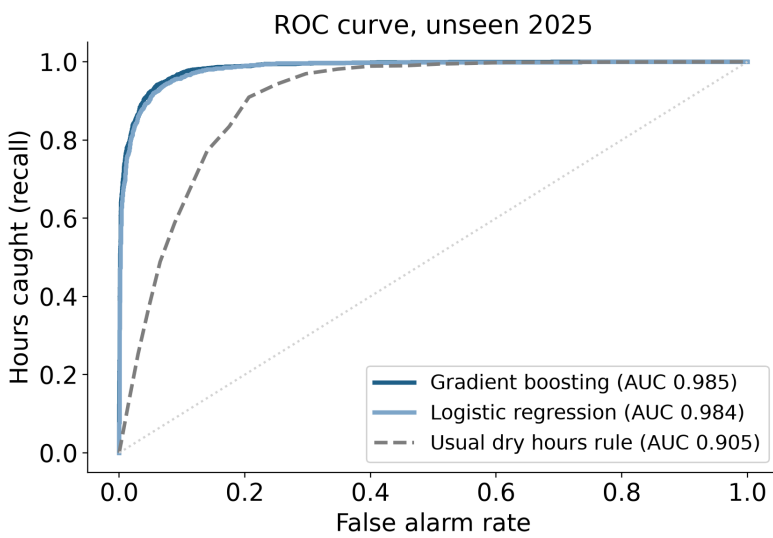
Soil moisture: drier topsoil means less evaporation and faster-rising VPD (Seneviratne et al., 2010).

ANALYSIS PROCESS

- Download raw CSV
- Clean
- Label: VPD > 1.2 kPa in 2 h
- Choose 29 features
- Split by time: train 2023–24, test 2025
- Baselines + logistic regression + gradient boosting
- Evaluate: F1, ROC, confusion
- Explain: top-5 features

One Jupyter notebook (Python, pandas, scikit-learn), runs in under a minute. "Always no" scores 76 % accuracy while catching nothing, so we judge on F1.

VISUALIZATIONS AND ROC CURVE



CONCLUSIONS

- The model beats every simple rule** on unseen 2025. It catches 87.2 % of misting-due hours, where "dry right now" catches 71.7 %, and its warnings are more precise (0.898 vs 0.839).
- Current VPD is the strongest signal.** The sun's daily cycle and soil moisture add the early warning.
- SDG 9:** an affordable, offline smart-farming tool that helps small growers act before plants are stressed.
- Challenge:** our first dataset had too few columns. We re-downloaded 25 variables and rebuilt the model in one notebook. **Lesson:** an honest time-split test beats a big score.
- Limits:** outdoor reanalysis data, not greenhouse air. **Next:** retrain on the rig's own logged readings.

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